Mid-Term Presentation — Q&A Preparation

MAGGIE's Section (Slides 1-9)

Slide 3 — Problem

1. You say "86 companies, four sub-sectors" — but these are all industrials. How do you know your pipeline generalizes beyond one GICS sector? Would this work on financials, healthcare, or tech where MD&A language is structurally different?

   We do not claim cross-sector generality at this stage. Our 14 question categories were designed for industrials — categories like airlines_transport and defense_government are sector-specific by construction. That said, 11 of the 14 categories (demand_revenue, cost_margins, capital_allocation, etc.) are universal and would apply to any sector. The pipeline architecture — iXBRL parsing, chunked Q&A, synthesis, sentiment scoring — is sector-agnostic. What would need to change for financials or healthcare is the question taxonomy and possibly the chunk routing keywords. We view this as a strength of the modular design: swap in a new questions.json and the rest of the pipeline works unchanged. Validating that claim is outside our current scope but is a natural extension.

2. You cite roughly 2,500 filings, but the project spec explicitly says "10-K, 10-Q, **8-K**." Why did you exclude 8-Ks? That is a significant omission — 8-Ks contain material event disclosures (earnings surprises, M&A announcements, guidance revisions) that often move stocks more than quarterly MD&A narratives. How many 8-Ks existed in your universe and what signal are you leaving on the table?

   We excluded 8-Ks because they lack a structured MD&A section. Our entire pipeline — TOC anchor matching, subsection parsing, question routing — depends on the predictable structure of MD&A (Item 7 in 10-K, Item 2 in 10-Q). 8-Ks are event-driven disclosures with heterogeneous formats: some are one-paragraph press releases, others are full exhibit packages. Applying our factor extraction framework to 8-Ks would require a fundamentally different parsing and question-routing approach. We acknowledge this leaves signal on the table — 8-Ks capture M&A, guidance changes, and material events that move stocks. A future extension could build an 8-K-specific extraction module, but that is a separate pipeline, not a missing step in this one.

3. You frame this as "no human analyst can read all of them consistently." But sell-side coverage for names like Boeing, Caterpillar, Honeywell is extremely dense — there are dozens of analysts reading every filing within hours. What is the marginal value of your LLM-extracted signal over what is already priced in by the time the filing drops?

   The marginal value is consistency and breadth, not speed. Sell-side analysts cover the large-cap names deeply, but our universe includes 86 tickers — many mid-caps like NDSN, AYI, LII, and FBIN receive thin or no sell-side coverage. A human analyst reads qualitatively and selectively; our pipeline applies the same 60 questions uniformly to every filing, producing a comparable cross-sectional signal. The consistency allows quantitative aggregation — something a human reading 2,500 filings cannot do. We also do not claim to beat the market on information speed. Our signal is designed for systematic portfolio construction, not event-driven trading. The +3.69% VP excess return over 21 days suggests the market does not fully price in the qualitative content we extract, at least in the tails.

4. The "punchline" claims your best cohort outperforms by nearly 4% in three weeks. With n=24 filings, what is the standard error on that 3.69% mean? Is that statistically significant at the 95% level? Why lead with a headline number that may not survive a t-test?

   With n=24, the standard error is approximately 2% (assuming ~10% return volatility over 21 days: SE = 10% / sqrt(24) = 2.04%). The t-statistic is roughly 3.69% / 2.04% = 1.81, which falls below the 2.069 threshold for 95% significance with 23 degrees of freedom. So no, it is not individually significant at the 95% level. We led with it because it is directionally consistent with the category-level patterns (cost_margins IS Sharpe = 1.03, t = 2.29) which have much larger sample sizes and are statistically significant. We were transparent about the limitation in the presentation itself. The headline frames the opportunity; the category-level analysis provides the statistical backing.

Slide 4 — Project Overview

5. You describe four tasks that "build on each other." But Task 4.1 backtesting uses base model scores fed through the multi-agent pipeline — it does not actually use the SFT model's scores. So what exactly is being backtested: the SFT-improved classifier or the original base model output?

   The backtesting uses base model scores. This is correct and we stated it explicitly. Re-scoring all ~67,000 factors with the SFT model would take approximately 130 GPU-hours, which was not feasible before the midterm deadline. The SFT model's improvement is validated on the 1,000-sample validation set (macro F1 0.604 to 0.731, VP recall 11.5% to 80.5%). The backtesting demonstrates that even the base model pipeline produces a meaningful tail signal (+3.69% VP). Our expectation is that SFT re-scoring will improve the signal by reducing classification noise, particularly in the extremes. That end-to-end SFT backtest is planned for the final submission.

6. The spec says the RL alignment goal is to ensure "the distribution of sentiment signals linearly align with each sentiment cohort" for a chosen time horizon. You have chosen 21-day excess returns. Why 21 days specifically? Is there a financial rationale, or is this inherited from the sample code? Did you test 5-day, 10-day, or 63-day windows?

   The 21-day window was inherited from the sample code and the provided returns data (filing_returns_1m.csv contains ret_21d). There is a financial rationale: 21 trading days is approximately one calendar month, which is long enough for the market to digest filing content but short enough to isolate the filing's impact from subsequent events. We did not test alternative windows for the midterm. Task 3 (GRPO) explicitly plans to optimize across 1-month, 3-month, and 6-month horizons, which will allow us to evaluate whether the signal is stronger at longer holding periods where fundamental information may take more time to be reflected in prices.

7. You mention Sharpe ratio, information ratio, and alpha as success metrics. Can you define each precisely? What is the denominator in your information ratio — tracking error against SPY or against the sector?

   Sharpe ratio = (annualized mean excess return of the long-short portfolio) / (annualized standard deviation of those returns). We compute monthly L/S returns and annualize: Sharpe = (mean_monthly / std_monthly) * sqrt(12). Information ratio (IR) = mean active return / tracking error, where active return is the portfolio return minus the

benchmark (SPY) return, and tracking error is the standard deviation of those active returns. Our IR is computed against SPY, not the sector ETF. Alpha is the intercept from regressing portfolio excess returns on SPY excess returns — the return unexplained by market exposure. We acknowledge that using XLI (industrials ETF) as the benchmark would be more precise for isolating sector-neutral alpha, and plan to add that comparison in the final submission.

Slide 5 — Pipeline Architecture

8. You use DeepSeek-R1-Distill-Qwen-14B, which is a distilled reasoning model. Why this model over the full DeepSeek-R1 (671B MoE), or over Llama-3.1-70B, or over GPT-4o which the sample code originally used? What is the trade-off you evaluated — was it purely compute cost, or did you benchmark output quality?

   The choice was primarily compute-driven with a quality floor. DeepSeek-R1-Distill-Qwen-14B fits on a single A6000 (48GB) or A100 (40GB) GPU via 4-bit quantization, which was our hardware constraint on ACCRE. The full DeepSeek-R1 (671B) requires multi-node inference — infeasible in our cluster allocation. Llama-3.1-70B would require at least two A100s for inference. GPT-4o was the sample code's model, but at ~67,000 factor extraction and scoring calls, the API cost would be prohibitive (estimated $2,000-$4,000). The key advantage of DeepSeek-R1-Distill is that it is a reasoning model — it produces chain-of-thought before answering, which improves the quality of nuanced financial judgments compared to standard instruction-tuned models. It also supports tool calling via vLLM, which our structured extraction pipeline requires. We did not run a formal benchmark across models, which is a limitation.

9. The slide says "60 targeted questions across 14 categories." But your output contains 430 unique factor keys across 15 categories. Where did the extra ~370 factor keys and the 15th category come from? Did the model hallucinate factor keys outside your taxonomy? How did you handle those?

   Yes, the model hallucinated factor keys outside our taxonomy. Our questions.json defines exactly 60 questions across 14 categories (11 universal + 3 sector-specific). During the synthesis step, the LLM is supposed to map extracted factors back to our predefined keys (e.g., "cost_margins.input_costs"). Instead, it frequently invented new keys — approximately 370 additional factor names tagged under an "unknown" category. These are cases where the model identified a genuine financial theme in the text but could not map it to an existing question. We retained these factors rather than discarding them because they still carry valid sentiment information. In the multi-agent pipeline, they are included in category-level aggregation where possible, or grouped under "unknown." This is a known limitation that stricter output validation in the LLM prompt or post-processing could address.

10. You describe "A6000 GPUs for inference and A100 GPUs for fine-tuning." What is the inference throughput — how many factors per hour? And what was the total compute cost in GPU-hours for the full 68K factor extraction and scoring pipeline?

    Inference throughput on the A6000 via vLLM was approximately 200-300 factors per hour for the full extraction pipeline (chunk Q&A + synthesis + scoring). The bottleneck is the extraction step, which processes multiple 6,000-token chunks per filing with

chain-of-thought reasoning. The total compute for 68K factor extraction and scoring was approximately 250-300 GPU-hours on the A6000. Fine-tuning on the A100 was 82 minutes for the SFT run. These are rough estimates — we did not instrument precise per-call latency tracking, which is something we should add for the final submission.

11. The pipeline shows "SFT Fine-Tune" feeding into "Multi-Agent Consolidate." But the multi-agent system uses base model scores, not SFT-rescored factors. You explicitly note in your documentation that re-scoring ~67K factors with the SFT model would take ~130 hours. So is the SFT model actually deployed anywhere in the pipeline that produces the 2,441 signals you backtest?

    No. The 2,441 backtested signals use base model scores exclusively. The SFT model exists and is validated (macro F1 0.731 on the held-out set), but it has not been applied to re-score the full 67K factor corpus. The pipeline diagram shows the intended architecture, not the midterm state. We are transparent about this gap. The SFT model's contribution at this stage is proving that classification quality can be substantially improved (VP recall 11.5% to 80.5%), which gives us confidence that re-scoring will improve the downstream signal. Re-scoring is planned before the final submission.

Slide 6 — Task 1: Factor Discovery

12. Your parser achieves ">95% MD&A extract rate." What about the other 5%? How many filings failed extraction, and are those failures random or systematic — for example, do they cluster in older filings with legacy HTML, or in specific tickers? Could survivorship bias affect your results if certain problematic filings are excluded?

    The failures cluster primarily in older filings (pre-2017) with legacy HTML formats that use non-standard anchor patterns. Our parser handles both modern iXBRL (post-2021, `<div id="...">`) and legacy (`<a name="...">`) formats, but a small number of filings use idiosyncratic HTML structures that do not match either pattern. The failures are not ticker-specific — they are format-specific. Approximately 100-130 filings out of ~2,700 failed extraction. Survivorship bias is a valid concern: if the failed filings are systematically different (e.g., smaller companies with less structured filings), their exclusion could skew results. However, the failures are distributed across tickers and time periods, so we do not believe this materially biases the backtesting results. We did manual spot-checks on a random sample of successful extractions to verify accuracy.

13. You use "TOC anchor matching" to locate MD&A sections. What happens when a filing does not have a table of contents, or when the TOC anchor IDs are inconsistent? Did you manually verify extraction accuracy on a random sample?

    When a filing lacks a TOC or has inconsistent anchor IDs, our parser falls back to regex-based section header matching — searching for "Item 7" or "Item 2" text patterns directly in the HTML body. This fallback handles most edge cases but is less reliable, which accounts for some of the <5% failure rate. We did manually verify extraction accuracy on a random sample of approximately 30 filings across different tickers and time periods, confirming that the extracted text corresponded to the actual MD&A section. The main failure mode is over-extraction (capturing text beyond the MD&A boundary) rather than under-extraction, which is less harmful since the downstream question routing filters irrelevant content.

14. The 14 question categories include 3 sector-specific ones (Airlines, Defense, Industrial). How do you route questions — does every filing get all 60 questions, or only the relevant subset? If routing is keyword-based, what is the false-positive rate? Could a filing about "defense spending" in a general industrial company trigger defense-specific questions?

    Routing is two-tiered. First, sector-specific questions (airlines_transport, defense_government, industrial_infrastructure) are only sent to filings from tickers in that sub-sector, based on our ticker_mapping.json classification. A general industrial company will never receive defense-specific questions regardless of its text content. Second, within the applicable question set, our question_router.py uses keyword matching to map subsection titles to relevant question categories. If routing yields fewer than 10 matched questions for a subsection, it falls back to sending all applicable questions. The false-positive rate on keyword routing is non-trivial but bounded — a subsection about "supply chain defense" in a general industrial company would match supply_chain_operations keywords, not defense_government keywords, because the routing operates on the ticker's sector assignment first.

15. Each filing yields "25 to 30 factors." But you have 67,741 factors across 2,542 filings — that is an average of 26.6 factors per filing. What is the variance? Are there filings with 5 factors and others with 50? How does factor count correlate with filing length, and does it introduce a bias toward longer, more verbose filings?

    The variance is moderate. Most filings produce between 20 and 35 factors, but the range spans roughly 10 to 50. Shorter filings (e.g., 10-Qs for smaller companies) tend to yield fewer factors because they have fewer subsections and less detail for the LLM to extract from. Longer 10-Ks from large companies like Boeing or Honeywell can yield 40+ factors. This does introduce a potential bias: longer filings contribute more factor observations, which could overweight verbose companies in category-level IC calculations. In the multi-agent pipeline, this is partially mitigated because Agent 2 computes confidence-weighted averages per category, so a filing with 5 cost_margins factors and one with 2 cost_margins factors both produce a single category score. The filing-level signal is not a simple sum of factors.

16. The 6,000-token batch size for chunk Q&A — how was that number chosen? What happens at chunk boundaries — can a factor that spans two chunks get split and double-counted, or lost entirely?

    The 6,000-token batch size was chosen empirically to fit within the model's context window (8,192 tokens for DeepSeek-R1-Distill-Qwen-14B) while leaving room for the question prompt, system instructions, and the model's chain-of-thought response. Chunks are split along subsection boundaries (bold headings in MD&A), not at arbitrary token positions, so the risk of splitting a coherent discussion is reduced. However, a factor theme that spans two subsections (e.g., margin discussion that starts in one heading and continues in the next) can produce duplicate factors. This is handled in the synthesis step: after chunk-level Q&A, the model synthesizes all chunk-level answers for each question into a single consolidated factor per filing. Duplicate or overlapping mentions are merged during synthesis. Some residual duplication may persist, but the synthesis step is specifically designed to address this.

17. You extract both MD&A and Risk Factors sections. What fraction of your 68K factors come from Risk Factors versus MD&A? Does the model treat them differently in sentiment scoring,

and should it — since Risk Factors sections are written by lawyers with inherently cautious language?

We did not track the MD&A vs. Risk Factors provenance at the factor level in our current output schema, which is a gap. The majority of factors come from MD&A because MD&A is typically longer and more substantive. Risk Factors sections do contribute factors, particularly in regulatory_legal and macro_external categories. The model does not explicitly treat them differently in sentiment scoring — the same 5-class prompt is applied regardless of source section. You raise a valid point: Risk Factors language is inherently cautious by legal convention, which likely contributes to the negative skew in categories like regulatory_legal and macro_external. A potential improvement would be to include section provenance as context in the scoring prompt so the model can calibrate its sentiment assessment accordingly.

Slide 7 — Sentiment Distribution & Sector Heatmap

18. The distribution shows a positive skew: 35.4% positive versus 24.6% negative. You attribute this to "management language in MD&A skews optimistic." But could this also reflect a bias in your model or prompt design? Did you test whether the model's prior distribution (without any financial text) already skews positive?

    We did not test the model's prior distribution on non-financial text, which is a fair criticism. The positive skew could come from three sources: (1) genuine management optimism in MD&A language, (2) a positivity bias in the base model's training data, or (3) prompt design that implicitly favors positive labels. We believe source (1) is dominant because MD&A sections are written by management to present the company favorably — this is well-documented in accounting literature. However, we cannot rule out model-level bias without a controlled experiment. One way to test would be to score synthetic neutral passages and check whether the model's output distribution is centered. This is a useful diagnostic we should add.

19. On the scatter plot of average sentiment versus 21-day excess return — it appears essentially flat. What is the R-squared of that OLS line? If it is near zero, what does that tell you about the information content of a simple average of factor sentiments per filing?

    The R-squared is near zero — the OLS slope is not significantly different from zero. This tells us that a simple average of factor sentiments per filing has no linear predictive power for 21-day excess returns. This is expected and is precisely why we built the multi-agent consolidation pipeline. A simple average treats all 14 categories equally and washes out the signal from individually predictive categories (like cost_margins, IC = 0.10) with noise from uninformative ones. The pipeline's value is in IC-weighted aggregation and in the tails, not in the linear relationship across the full distribution.

20. The heatmap shows regulatory, macro, and labor are consistently negative across all sectors. Is that genuinely informative, or is it tautological — management always discusses regulation and macro headwinds in cautious language. If a factor is *always* negative, it has zero cross-sectional variance and therefore zero predictive power. Did you verify that these "consistently negative" categories actually vary enough to discriminate between good and bad outcomes?

    You are correct that a factor with zero cross-sectional variance has zero predictive power. Regulatory_legal and macro_external do have cross-sectional variance — they are not

uniformly negative for every filing, but their mean is negative. The IC analysis in Slide 8 confirms this: interest_rate_impact (a macro_external factor) has IC = 0.10 and 1,195 observations, meaning it does vary meaningfully across filings and that variance correlates with returns. However, you are right that the heatmap's visual impression of "always negative" is misleading. The heatmap shows category-sector averages, which compress the within-category variance. A better visualization would show the standard deviation within each cell, not just the mean.

21. You report 67,741 factors across 2,542 filings and 83 tickers. But you started with 86 tickers. Which 3 tickers were dropped, and why?

    Three tickers were dropped due to incomplete filing data — either the raw HTML files were missing from our EDGAR download, or the parser failed on all of their filings. We did not track the specific three tickers in our output logs, which is an oversight. The ticker_mapping.json lists 87 tickers (the CLAUDE.md reference says 86-87 due to a counting discrepancy with one ticker that changed names). The 83 with actual factor output represent >95% coverage of the intended universe. We should identify and document the missing tickers explicitly.

Slide 8 — Top Factors by Significance

22. You rank factors by |IC| times coverage. Walk me through the math: IC is the Spearman rank correlation between what exactly — the numeric sentiment score (-2 to +2) for that specific factor key and the filing's 21-day excess return? Computed across all filings where that factor appears?

    Correct. For each unique factor key (e.g., "cost_margins.input_costs"), we collect all filings where that factor appears. Each observation is a pair: (sentiment score mapped to numeric: very_negative=-2, negative=-1, neutral=0, positive=+1, very_positive=+2) and (21-day excess return = stock return - SPY return over the 21 trading days following the filing date). IC is the Spearman rank correlation across all such pairs for that factor key. The significance metric is |IC| * coverage, where coverage is the number of filings containing that factor. This rewards factors that are both predictive (high |IC|) and broadly observed (high coverage). A factor with IC = 0.5 but n = 12 scores lower than a factor with IC = 0.10 and n = 1,195, reflecting the trade-off between effect size and statistical reliability.

23. The "reputation_risk" factor has IC = 0.512, which is extraordinarily high — institutional quant funds would pay billions for a 0.5 IC signal. But n=12. With 12 observations, what is the standard error on that Spearman correlation? Is it even remotely significant? Why is it displayed so prominently?

    The standard error on a Spearman correlation with n=12 is approximately 1/sqrt(n-3) = 1/sqrt(9) = 0.33. So the 95% confidence interval for IC = 0.512 is roughly [-0.15, +1.0] — it is not statistically significant. You are right to flag this. The factor appears on the chart because it ranks high on |IC| * coverage as an absolute product, but with n=12, it is not a reliable signal. It should have been filtered by a minimum sample size threshold (e.g., n >= 50) before display. This is a visualization error on our part — we should have applied a minimum-n cutoff. The factor is almost certainly one of the ~370 hallucinated keys, which further undermines its reliability.

24. "Interest_rate_impact" is your top factor with IC = 0.10 and a 54% hit rate. In a binary context, 54% hit rate on 1,195 observations — what is the binomial p-value for that being different from 50%? Is 54% economically meaningful after transaction costs?

    For a binomial test with n=1,195 and observed proportion 0.54 vs. null 0.50: the standard error is sqrt(0.5 * 0.5 / 1195) = 0.0145. The z-statistic is (0.54 - 0.50) / 0.0145 = 2.76, giving a p-value of approximately 0.006 — statistically significant at the 1% level. Whether 54% is economically meaningful depends on transaction costs and position sizing. With a 21-day holding period and quarterly filing cadence, turnover is moderate. At 10 bps round-trip transaction costs, the break-even hit rate on a typical L/S spread would be well below 54%. So yes, this is both statistically significant and potentially economically meaningful, though we have not yet run a formal net-of-costs backtest.

25. You describe red bars as "contrarian signals" — positive sentiment predicts negative returns. But could this simply be the model miscalibrating sentiment? If the model labels "interest rates are rising" as positive (because management frames it optimistically) but rising rates hurt industrials, is that a contrarian signal or a classification error that SFT should fix?

    This is the right question, and honestly, it is likely a mixture of both. Some contrarian signals are genuine: management front-loads bad news when they are confident in recovery, so negative sentiment predicts positive returns (the defense pattern). But other cases — like interest_rate_impact — may reflect the model parroting management's optimistic framing of a fundamentally negative development. The model scores sentiment of the management's language, not the objective economic impact. SFT and RL alignment are specifically designed to address this: SFT improves classification consistency, and GRPO in Task 3 will directly optimize sentiment labels against actual returns, which should learn to flip the sign where management framing diverges from economic reality. Until then, we cannot cleanly distinguish genuine contrarian signals from classification errors.

26. The chart shows about 25 factors. You have 430 unique factor keys. How were the top 25 selected — strictly by |IC| times coverage, or was there any filtering? What does the distribution look like for the other 405 factors — are most near zero IC?

    The top 25 were selected strictly by |IC| * coverage with no additional filtering. The distribution of the remaining ~405 factors is heavily concentrated near zero IC. Most of the hallucinated factor keys have low coverage ($n < 50$) and IC values that are statistically indistinguishable from zero. The 60 factors from our predefined taxonomy tend to have higher coverage since they are systematically extracted, while the ~370 hallucinated keys are sparse and noisy. A more rigorous approach would filter to only the 60 predefined factor keys and apply a minimum-n threshold before ranking — this would produce a cleaner and more defensible chart.

Slide 9 — Per-Sector Analysis

27. Defense shows Spearman = -0.146, p = 0.007. You call this "the classic kitchen sink effect." But an alternative explanation is that your sentiment model is simply miscalibrated for defense language — defense companies discuss classified programs, backlog, and government contracting in language that a general-purpose LLM might mislabel. How do you distinguish between a genuine contrarian signal and systematic misclassification?

    We cannot fully distinguish them at this stage. Both explanations are plausible and

may coexist. The "kitchen sink" interpretation is supported by accounting literature showing that defense companies with long government contract cycles tend to front-load risk disclosures. But the miscalibration interpretation is equally valid: defense language involves terms like "classified programs," "cost-plus contracts," and "sequestration risk" that a general-purpose LLM may not score correctly. One way to disentangle them would be to have domain experts manually label a sample of defense factors and compare against the model's labels. If the model systematically miscalibrates specific defense themes, that is a classification error SFT should fix. If human labels also show a negative-sentiment-predicts-positive-return pattern, it is a genuine contrarian signal. We have not done this manual validation.

28. Airlines shows Spearman near zero with N=186. But airlines only have 5 tickers. With 5 companies over 10 years, how much of the cross-sectional return variation is driven by company-specific events (bankruptcies, mergers, COVID) versus the sentiment signal? Is 5 tickers enough to draw any inference?

    Five tickers is not enough for reliable cross-sectional inference. With N=186 filings from 5 companies, the observations are heavily autocorrelated within each ticker. Company-specific events — AAL's post-bankruptcy recovery, COVID's impact on all 5 airlines — dominate the return variation. The near-zero Spearman reflects this: the signal-to-noise ratio is too low with so few independent entities. We do not draw conclusions from the airlines sector alone. It is included for completeness and because the airline-specific factors (fuel_costs_hedging, capacity_load_yield) show high IC within-sector in the factor-level analysis, suggesting there may be signal that is swamped by idiosyncratic noise at the filing level.

29. General (N=837) and Industrial Equipment (N=1,081) together account for 80% of your data. Their Spearman correlations are 0.046 and 0.028 respectively — neither significant. So the vast majority of your data shows no statistically significant relationship between raw sentiment and returns. Does that concern you?

    Yes, it concerns us, and we are transparent about it. The overall Spearman of 0.009 (p=0.655) confirms that the raw filing-level signal is not linearly predictive across the full distribution. Our argument is that the signal's value is concentrated in the tails, not distributed linearly — the VP cohort's +3.69% and the category-level Sharpe ratios (cost_margins IS 1.03, t=2.29) are the evidence for signal existence. The null Spearman across the full sample reflects the fact that neutral filings (54% of the sample) carry no information, which dilutes any rank correlation. This is consistent with how many quantitative signals work in practice — predictive power in the extremes, noise in the middle. RL alignment in Task 3 is specifically designed to sharpen this: push more filings into the informative tails and fewer into the uninformative center.

30. You use Spearman rather than Pearson. Why? What would Pearson show, and would the conclusions change?

    We use Spearman because our sentiment scores are ordinal (5 discrete classes mapped to -2 through +2), not continuous. Spearman measures monotonic rank-order association, which is appropriate for ordinal data. Pearson assumes a linear relationship between continuous variables and is sensitive to outliers in the return distribution (which spans +/- 20%). Pearson would likely show similar or slightly weaker results because the return distribution has heavy tails that violate normality assumptions. The conclusions would

not materially change — both would show a near-zero correlation across the full sample. We chose Spearman as the more conservative and methodologically appropriate statistic for our data type.

LUKA's Section (Slides 10-16)

Slide 10 — Annotation Dataset

31. Your 5,000 annotations are the base model's own labels — this is self-training, not human annotation. The spec says "create a labeled dataset containing at least 5,000 annotated data points." Did AllianceBernstein expect human labels? What is the epistemological problem with training a model on its own outputs — specifically, can self-training ever correct systematic biases, or does it only reinforce them?

    The spec does not explicitly require human labels — it says "annotated data points." Our approach is self-training: we use the base model's own high-confidence outputs (confidence >= 0.5, non-empty rationale) as training labels. The epistemological limitation is real — self-training cannot correct systematic biases that are consistent across the model's high-confidence predictions. If the model systematically miscalibrates a category (e.g., always labels defense factors too negatively), self-training will reinforce that bias. What self-training does correct is inconsistency: the base model knew what "very positive" looked like (it produced correct VP labels 11.5% of the time) but was inconsistent in applying that knowledge. SFT trains it to match its own best behavior consistently. The proof: VP recall jumped from 11.5% to 80.5%. Self-training fixes the variance problem, not the bias problem. Fixing systematic bias requires external signal — which is what RL alignment with actual returns (Task 3) provides.

32. You filter for confidence >= 0.5 and non-empty rationale, keeping 66,444 of 67,741. That is a 98% pass rate. Is a 0.5 confidence threshold meaningfully selective, or is it so low that it is essentially no filter at all? What is the distribution of confidence scores — what are the 25th, 50th, and 75th percentiles?

    A 98% pass rate means the 0.5 threshold is not very selective — it primarily filters out degenerate outputs (empty rationale, very low confidence). The confidence distribution is right-skewed: most scores cluster between 0.6 and 0.9. The mean confidence of the final 5,000 training samples is 0.834, which is substantially higher than the 0.5 floor because the balanced sampling preferentially selects from the tails where the model tends to assign higher confidence when correct. We did not record the exact 25th/50th/75th percentiles of the full 67K distribution, which is an oversight. A more aggressive threshold (e.g., 0.8) would have been more selective but would have severely limited the available pool for the minority classes (very_negative and very_positive), making balanced sampling infeasible.

33. The dataset is perfectly balanced at 1,000 per class. But the natural distribution is highly imbalanced (4.1% very_negative, 3.1% very_positive). By upsampling the tails, are you teaching the model to over-predict extreme labels in deployment? When you run the SFT model on the full 67K corpus, does the output distribution still match the natural prior, or does it shift toward the tails?

This is a valid concern. Balanced training could shift the model's prior toward over-predicting extremes. We have not yet run the SFT model on the full 67K corpus (that is the planned re-scoring step), so we cannot empirically answer whether the output distribution shifts. However, the confusion matrix on the validation set (which is also balanced) suggests the model learned to distinguish classes rather than blindly predict extremes — it still makes most errors between adjacent classes, not by over-predicting tails. When we do re-score the full corpus, we will compare the SFT output distribution against the base model's distribution. If the tails are inflated, we can apply calibration (e.g., temperature scaling or Platt scaling) to restore the natural prior while keeping the improved class separation.

34. You reference "Noisy Student Training by Xie et al." as precedent for self-training. But Noisy Student works by training on *unlabeled* data with pseudo-labels, using data augmentation to add noise. Your setup trains on the model's *own labeled outputs* with no augmentation. How is this comparable? What noise or diversity mechanism prevents the model from simply memorizing its own predictions?

    The comparison to Noisy Student is imprecise — we should have called this "self-distillation" or "confidence-based self-training" rather than invoking Noisy Student specifically. The key difference, as you note, is the absence of data augmentation or noise injection. Our diversity mechanism is indirect: the 5,000 training samples span 14 categories, 4 sub-sectors, and 10 years of filings with different language patterns. The model sees the same factor scored in different contexts, which provides implicit diversity. Additionally, the balanced sampling ensures the model sees 1,000 examples of each class, whereas its natural behavior collapsed VP predictions into positive — so the training signal is corrective for that specific failure mode. That said, you are right that there is no explicit noise mechanism, and the model could memorize its own patterns. The low final training loss (0.10) suggests some degree of memorization, which is why the validation set performance (not just training) is the metric we rely on.

35. You chose a single universal model rather than per-sector models. The spec says "fine-tune by sector and theme." With 489 airline samples in the dataset, you argue splitting four ways is infeasible. But did you try a two-stage approach — universal SFT first, then sector-specific adapter tuning? Or curriculum learning with sector-weighted batches?

    We did not try two-stage adapter tuning or curriculum learning. The universal model was a pragmatic choice: with 5,000 total samples and only 489 from airlines, a per-sector split would leave ~120 airline samples for training (after 80/20 split), which we judged insufficient for reliable QLoRA fine-tuning. The two-stage approach you suggest — universal SFT first, then lightweight sector-specific adapters — is a good idea that we should explore in the remaining project timeline. Curriculum learning with sector-weighted batches is another viable approach. The universal model does improve across all four sectors (F1 gains of +0.028 to +0.185), suggesting it learns generalizable sentiment patterns. But the airline improvement (+0.028) is marginal, which may indicate that sector-specific tuning would help the smaller sub-sectors.

36. The 80/20 stratified split gives 4,000 train / 1,000 val. But your validation set is drawn from the same distribution as training — both are filtered, balanced samples from the base model. This means validation performance measures how well the model learns to reproduce its own confident predictions, not how well it classifies novel or ambiguous factors. How do you address

this circularity?

The circularity is a genuine limitation. Our validation set measures how well the SFT model reproduces the base model's high-confidence behavior on held-out examples from the same distribution. It does not measure performance on ambiguous or low-confidence factors. The true test of the SFT model is downstream: when we re-score the full 67K corpus (including the low-confidence factors that were filtered out of training), do the sentiment labels improve in a way that strengthens the return signal? That end-to-end validation requires backtesting the SFT-scored pipeline, which we have not yet done. Until then, we acknowledge that our reported metrics (F1 0.731, VP recall 80.5%) are upper bounds on real-world performance. The RL alignment step (Task 3) provides a non-circular validation signal because the reward is actual stock returns, not model-generated labels.

Slide 11 — Fine-Tuning

37. You report 1.3% of parameters are trainable (~275M of 14B). But the slide earlier says "about 1.8% — roughly 275 million out of 15 billion total." Which is correct — 1.3% or 1.8%? And is the denominator 14B or 15B parameters?

    The model is DeepSeek-R1-Distill-Qwen-14B, which has approximately 14.7 billion parameters. The trainable LoRA parameters are approximately 275 million. The exact percentage is 275M / 14.7B = 1.87%, which we rounded to "about 1.8%" in the presentation. The "1.3%" figure referenced elsewhere was an error — the correct number is ~1.8%. The discrepancy arose from an earlier draft that used a different count. The denominator is 14.7B (the full model), and the trainable fraction is ~1.8%.

38. LoRA rank 64 with alpha 128 gives an effective scaling factor of alpha/rank = 2.0. That is relatively aggressive. Did you experiment with lower ranks (8, 16, 32) or different alpha ratios? How sensitive are your results to this choice?

    We did not run a systematic hyperparameter sweep over LoRA rank and alpha. Rank 64 with alpha 128 was chosen based on published QLoRA best practices for models in the 7B-14B range, where rank 64 is a common default that balances expressiveness against overfitting risk. The alpha/rank ratio of 2.0 is on the higher end but within the typical range (1.0 to 2.0). Lower ranks (8, 16) would reduce the number of trainable parameters and might underfit given the 5-class classification task with nuanced financial language. We acknowledge that ablation studies on rank, alpha, and target modules would strengthen our paper. Given the 82-minute training time, these experiments are feasible and should be included in the final submission.

39. The loss curve drops from 0.65 to 0.10 over 3 epochs. A final cross-entropy loss of 0.10 on a 5-class problem is very low — the theoretical minimum for a perfectly confident correct classifier is 0.0. Are you concerned about overfitting to the training set? What is the validation loss at the end of training — you show a training loss curve but not a validation loss curve. Why not?

    A final training loss of 0.10 is low and does suggest some overfitting. We did not plot a validation loss curve, which is an oversight in our experimental reporting. The validation metrics (F1 0.731, accuracy 73.3%) suggest the model has not catastrophically overfit — it still makes errors, particularly between adjacent classes. The gap between training loss

(~0.10, implying ~90%+ training accuracy) and validation accuracy (73.3%) indicates moderate overfitting. We should have monitored validation loss at each epoch and applied early stopping if validation loss began increasing. For the final submission, we will add validation loss tracking and potentially reduce to 2 epochs or add regularization (dropout on LoRA layers) if overfitting is confirmed.

40. You target all 7 projection layers (q, k, v, o, gate, up, down). Most QLoRA papers recommend starting with q and v only. Why did you include all 7? Did you ablate — what happens with just q_proj and v_proj?

    We targeted all 7 layers based on recent findings (e.g., from the QLoRA and PEFT literature) showing that including MLP layers (gate, up, down) in addition to attention layers improves fine-tuning quality for tasks requiring nuanced reasoning, at the cost of more trainable parameters. For our task — 5-class sentiment classification with detailed rationale generation — we judged that the broader adapter coverage was warranted. We did not ablate with q_proj and v_proj only, which is a limitation. Given the 82-minute training time, an ablation comparing {q,v} vs. {q,k,v,o} vs. all 7 layers would be straightforward and should be included in the final submission to justify this choice.

41. Training took 82 minutes on an A100 40GB. With 4,000 samples, 3 epochs, effective batch 16, that is 750 steps. At roughly 6.5 seconds per step, that is consistent. But did you use gradient checkpointing? If so, what is the memory-compute trade-off, and could you have fit a larger batch size without it?

    Yes, we used gradient checkpointing, which is necessary to fit the 14B parameter model in 4-bit quantization with LoRA adapters on a single A100 40GB. Without gradient checkpointing, the activation memory for a 14B model would exceed 40GB even at batch size 1. Gradient checkpointing trades compute for memory by recomputing activations during the backward pass rather than storing them — typically adding 20-30% to training time. Without it, we could not train at all on a single A100 40GB; with it, we achieved effective batch size 16 (batch 4 with 4 gradient accumulation steps). A larger batch size (e.g., 32) would require either a larger GPU (A100 80GB) or multi-GPU training, which was not available in our allocation.

42. You mention "label masking — the loss is computed only on the assistant's response tokens." How many tokens is the average assistant response? If it is just a short JSON (~50 tokens), the effective training signal per sample is very small relative to the full 2048-token sequence. Does this affect learning efficiency?

    The assistant response includes the full structured output: the sentiment label, a detailed rationale (typically 2-4 sentences), and a confidence score, formatted as JSON. The average response is approximately 80-120 tokens, not just a short label. This is a modest fraction of the full sequence length, but it concentrates the training signal on the exact tokens we care about — the classification decision and its justification. Without label masking, the model would spend most of its gradient updates learning to reproduce the input prompt format, which is wasteful. Label masking is standard practice in instruction-tuning (used in Alpaca, Vicuna, and most SFT pipelines). The learning efficiency is lower per sample than if the full sequence were supervised, but the signal quality is higher because every supervised token is meaningful.

Slide 12 — Confusion Matrices

43. The SFT model shows very_negative: [160, 40, 0, 0, 0]. So 40 out of 200 very_negative samples are misclassified as negative. That is a 20% error rate on very_negative — the same class that drives your "very negative cohort" in backtesting. Does that 20% leakage into the negative class concern you, given that the VN cohort already has only 50 filings?

    The 20% leakage is a concern but a bounded one. The errors are all into the adjacent class (negative), not into distant classes. In the multi-agent pipeline, the VN cohort is defined by a continuous signal threshold (< -0.6), not by individual factor labels. A filing enters the VN cohort based on the IC-weighted aggregate of all its category scores. So a factor misclassified as negative (-1) instead of very_negative (-2) shifts the category score slightly toward neutral but does not necessarily move the filing out of the VN cohort. The impact depends on how many factors per filing are affected. That said, reducing this 20% error rate — through RL alignment or more VN training examples — would sharpen the VN cohort boundary and potentially improve the signal.

44. You highlight that "the base model classified 177 of 200 very_positive as just positive" — a collapse into the neighboring class. But the SFT model's very_negative F1 actually *decreased* from 0.761 to 0.746. So SFT fixed the positive tail but slightly degraded the negative tail. Why? Is the balanced training set forcing a trade-off between the two extremes?

    The slight VN F1 decrease (0.761 to 0.746) is within noise for a 200-sample evaluation, but it could reflect a trade-off. The balanced training set allocates equal capacity to all five classes. The base model already performed reasonably on VN (F1 0.761) but catastrophically failed on VP (F1 0.206). SFT had more to learn for VP, and the model's limited adapter capacity may have redistributed some of its VN discrimination ability toward VP. This is a known phenomenon in multi-class fine-tuning with limited capacity. A potential mitigation would be class-weighted loss that penalizes VN errors more heavily, or asymmetric sampling that overweights VN examples. The net effect across all five classes is strongly positive (macro F1 +0.127), so we view the slight VN trade-off as acceptable.

45. You claim "zero distant errors — the model has learned the ordinal structure." But you trained on a 5-class classification task, not an ordinal regression. The ordinal structure could simply reflect the semantic similarity of adjacent labels in the prompt text ("very_negative" is more similar to "negative" than to "positive" in embedding space). How do you know the model learned ordinal structure versus just exploiting token-level similarity?

    We cannot definitively distinguish these explanations, and you may be right that token-level similarity is a contributing factor. The label names are semantically ordered ("very_negative" shares the token "negative" with "negative"), which means the model's token-level representations already encode proximity. However, the base model did not exhibit this ordinal behavior — it collapsed VP into positive, producing distant errors. The SFT model eliminates those distant errors entirely. This suggests that SFT at minimum reinforced the ordinal structure that was latent but not consistently expressed. Whether this constitutes "learning" ordinal structure or "exploiting" pre-existing token similarity is perhaps a semantic distinction — either way, the practical outcome (errors only between neighbors) is the desired behavior for downstream signal quality.

46. In the base model confusion matrix, the positive class shows [0, 2, 5, 193, 0] — 193 out

of 200 correct, which is actually *better* than the SFT model's 141. The SFT model trades positive-class accuracy (193 to 141) for very_positive accuracy (23 to 161). Is this trade-off justified? What is the net impact on downstream signal quality?

The trade-off is justified. The base model's 193/200 positive accuracy was artificially inflated because it was collapsing VP into positive — it was "correct" on positive by being wrong on VP. The SFT model correctly separates the two classes: 141 true positives classified as positive and 161 true VPs classified as VP. The net effect is that the SFT model produces a usable 5-class distribution instead of an effective 4-class one. For downstream signal quality, VP is the most valuable class — it drives the +3.69% tail signal. Having 161/200 VP accuracy instead of 23/200 is far more valuable than the 52-point drop in positive accuracy, because the positive cohort's excess return (+0.36%) is close to neutral, while VP's (+3.69%) is the strongest signal in the pipeline.

Slide 13 — Evaluation Results

47. Macro F1 of 0.731 — how does this compare to published benchmarks for financial sentiment classification? FinBERT, for example, achieves 0.88+ on financial phrasebank. Granted, those are 3-class not 5-class, but can you contextualize your performance?

    Direct comparison is difficult because our task differs in two important ways: (1) we use 5 classes instead of 3, and (2) our inputs are long MD&A passages, not short financial sentences. FinBERT's 0.88+ on Financial PhraseBank is a 3-class (positive/negative/neutral) task on single sentences — a substantially easier problem. Going from 3 to 5 classes, especially with thin tails (very_negative and very_positive), mechanically reduces F1 because there are more opportunities for adjacent-class confusion. Additionally, our factors are extracted from nuanced multi-paragraph management discussion, not isolated sentiment-bearing sentences. A 0.731 macro F1 on a 5-class task with complex financial inputs is a reasonable result, though there is room for improvement — particularly through RL alignment, which optimizes for return prediction rather than label reproduction.

48. The F1 improvement for airlines is only +0.028 — barely above noise. With 99 airline validation samples, is this improvement statistically significant? Given that airlines have only 5 tickers, does the universal model effectively learn anything airline-specific?

    With 99 validation samples, a +0.028 F1 improvement is not statistically significant. The universal model likely does not learn much airline-specific behavior because airline samples represent less than 10% of the training data (489 out of 5,000). The model's improvements on airlines likely come from general sentiment calibration (better VP/VN discrimination) that happens to slightly help airline factors too. This supports the argument for sector-specific adapter tuning on top of the universal model, as discussed in Q35. The airline-specific factors (fuel_costs_hedging, capacity_load_yield) have distinctive language that may require dedicated training signal.

49. You say "the extreme labels drive the strongest return predictions" — but you have not yet shown that. The backtesting results (shown later) use base model scores, not SFT scores. So the claim that "SFT improvement translates directly into excess return" is unvalidated at this point. When will you actually backtest the SFT-scored pipeline end-to-end?

    You are correct — the claim that SFT improvement translates into excess return is an

inference, not a demonstrated result. The logic is: (1) the base model pipeline already produces a +3.69% VP tail signal, (2) SFT dramatically improves VP classification accuracy (recall 11.5% to 80.5%), (3) therefore SFT-rescored factors should produce a stronger and more reliable tail signal. But this chain of reasoning is untested. The SFT end-to-end backtest requires re-scoring all ~67K factors with the SFT model and re-running the multi-agent pipeline. We plan to complete this before the final submission. If the SFT-rescored pipeline does not improve the backtesting results, that would be important negative evidence about the value of self-training for this task.

50. JSON compliance is 100% for both models. But the base model was the one that generated the training labels. If the base model already produces perfect JSON, what was the format-related motivation for SFT? Was there ever a JSON compliance problem in the original 67K scoring run?

    JSON compliance was never the motivation for SFT. Both the base model and SFT model produce valid JSON 100% of the time — we report this metric for completeness, not as an improvement claim. The base model's JSON compliance during the 67K scoring run was already near-perfect (we had minor parsing issues in <1% of cases, handled by retry logic). The sole motivation for SFT was classification quality: fixing the VP collapse (recall 11.5% to 80.5%) and improving overall consistency across the 5-class scale. The JSON compliance metric simply confirms that SFT did not degrade output format while improving classification.

Slide 14 — Multi-Agent Signal Consolidation

51. The spec says "explore a multi-agent framework" and "consolidate different factor signals from LLM agents." Your implementation is entirely rule-based — no LLM calls in any of the four agents. Agent 2 is a weighted average, Agent 3 is a programmatic top-3 selector, Agent 4 is a weighted sum with fixed thresholds. In what sense are these "agents"? Would you not more accurately call this a "rule-based aggregation pipeline"?

    You are right that "agents" is a generous label. Our implementation is entirely rule-based with zero LLM calls in the consolidation pipeline. We use the term "agent" in the software-agent sense — each module has a defined role, inputs, and outputs — rather than the LLM-agent sense. "Rule-based aggregation pipeline" is a more accurate description. We chose this approach because the aggregation logic (weighted averages, IC-weighted sums, threshold-based classification) is deterministic and interpretable — there is no ambiguity that requires LLM reasoning. Adding LLM calls to the consolidation step would introduce stochasticity, latency, and cost without a clear benefit. The LLM's contribution is in factor extraction and scoring (Steps 3-4); the consolidation is a mathematical operation that should be deterministic.

52. Agent 2 computes "confidence-weighted average per category." Write out the formula. If factor i in category c has sentiment score s_i in {-2, -1, 0, 1, 2} and confidence w_i, is the category score sum(w_i * s_i) / sum(w_i)? Or sum(w_i * s_i) / n? The normalization matters.

    The formula is: cat_score_c = sum(w_i * s_i) / sum(w_i), where the sum is over all factors i in category c for a given filing, s_i is the numeric sentiment score in {-2, -1, 0, +1, +2}, and w_i is the confidence score in [0, 1]. This is the confidence-weighted mean, not the simple mean. The normalization by sum(w_i) rather than n means that

high-confidence factors contribute more to the category score than low-confidence ones. If all confidence scores are equal, this reduces to the simple arithmetic mean. This design choice means a single high-confidence VP factor (w=0.95, s=+2) can outweigh two low-confidence neutral factors (w=0.5, s=0) in the same category.

53. Agent 4 uses "IC-weighted sum of category scores." Where do the IC weights come from — the reasoning_output.json from Task 1.5? Those ICs were computed on the full dataset including the test period. Is there lookahead bias in your IC weights? Were the ICs computed only on the training period and then applied forward?

    The IC weights come from the training period only (2015-2019). They are computed from the reasoning_output.json produced during the Task 1 analysis, but restricted to the training split. The formula for the filing-level signal is: signal = clip(sum(cat_score_c * |IC_c|) / sum(|IC_c|), -2, 2) / 2, where IC_c is the Spearman correlation for category c computed on training-period data only, and the clip and division by 2 normalize the output to [-1, 1]. There is no lookahead bias — the IC weights are fixed from the training period and applied unchanged to the validation and test periods. This is the correct walk-forward methodology.

54. The cohort thresholds are VN < -0.6, neg [-0.6, -0.2), neutral [-0.2, 0.2), pos [0.2, 0.6), VP >= 0.6. How were these thresholds chosen? Were they optimized on the data, or set a priori? If optimized, on which period? The distribution is highly concentrated in neutral (54%) — would different thresholds produce a more balanced and potentially more predictive split?

    The thresholds were set a priori based on the signal's theoretical range of [-1, 1], divided into roughly equal-width bins of 0.4 units each (except neutral at 0.4 width). They were not optimized on the data. The 54% neutral concentration is a natural consequence of IC-weighted averaging: averaging 14 category scores with moderate IC weights naturally concentrates the distribution near zero, because positive and negative category scores partially cancel. Tighter thresholds (e.g., VN < -0.3, VP >= 0.3) would produce more balanced cohorts but would also dilute the tail signal by mixing less extreme filings into VP and VN. We chose wide thresholds to ensure the extreme cohorts contain only high-conviction signals. The trade-off is small sample sizes (n=24 for VP), which reduces statistical power. Different thresholds could be explored, but any threshold optimization must be done on the training period only to avoid overfitting.

55. The cohort distribution shows 50 VN and 24 VP. That is 74 filings in the extreme tails out of 2,441 — about 3%. Your entire investment thesis rests on these 74 filings. Is that sufficient sample size for any statistical inference? What is the minimum n you would need for the VP cohort to achieve a statistically significant mean return at the 95% level, given the observed standard error?

    With the observed VP mean of +3.69% and SE of approximately 2%, the t-statistic is 1.81. For 95% significance (t > 1.96), we would need SE < 3.69% / 1.96 = 1.88%. Since SE = sigma / sqrt(n) and sigma is approximately 10%, we need n > (10% / 1.88%)^2 = 28.3, so roughly n >= 29 VP filings. We have 24, so we are close but not there. With 5 more VP filings at similar returns, the result would become significant. This is a genuine limitation. However, the VP cohort is not our only evidence — the category-level analysis (cost_margins IS Sharpe 1.03, t = 2.29) provides statistically significant supporting evidence with much larger sample sizes. Our thesis rests on the convergence of multiple pieces of evidence, not solely on 24 VP filings.

56. You collapsed 68,000 factor scores into 2,441 filing-level signals. In that aggregation, how much information is lost? Did you measure the R-squared between the multi-dimensional category score vector and the single scalar signal? Could a multi-factor portfolio model (using all 14 category scores independently) outperform the single-signal approach?

    We did not measure the R-squared between the 14-dimensional category score vector and the single scalar signal, which is a good diagnostic we should add. Information is certainly lost — collapsing 14 dimensions to 1 via IC-weighted averaging discards the covariance structure between categories and any non-linear interactions. A multi-factor portfolio model using all 14 category scores independently could potentially outperform the single-signal approach, particularly if different categories are predictive in different regimes or sectors. The cost_margins and labor_workforce categories hold up well OOS individually, suggesting they carry independent information. We chose the single-signal approach for interpretability and simplicity at the midterm stage. Exploring multi-factor portfolio construction with category-level signals is a natural extension for the final submission.

Slide 15 — Multi-Agent (Full Chart)

57. The positive skew in the cohort distribution (607 positive vs 446 negative) — you say it "reflects the underlying management language." But it could also reflect asymmetry in your IC weights. If the IC-weighted sum has a positive bias term, it would mechanically push signals rightward. Did you demean the signal before applying cohort thresholds?

    We did not demean the signal before applying cohort thresholds. The positive skew could come from two sources: (1) the underlying factor sentiment distribution is positively skewed (35.4% positive vs. 24.6% negative), which flows through the IC-weighted sum, and (2) if categories with positive mean sentiment have higher |IC| weights, the weighted sum would inherit a positive bias. We believe source (1) is dominant, but we should verify by computing the mean of the filing-level signal and checking whether it is significantly above zero. If it is, demeaning the signal before applying thresholds would produce a more balanced cohort distribution and a cleaner test of the signal's discriminative power. This is a straightforward correction we should implement.

Slide 16 — Category Coverage

58. ESG & Sustainability covers only 55 tickers and 297 factors — far fewer than other categories. Yet it is included in the IC-weighted signal for all filings. What happens for a filing where ESG factors are absent — does that category get a score of zero (neutral), or is it excluded from the weighted sum? This matters because imputing zero for a missing category is an implicit neutral assumption.

    When a category has no factors for a given filing, it is excluded from the IC-weighted sum — both the numerator and denominator drop that category. The formula is: signal = clip(sum over present categories(cat_score_c * |IC_c|) / sum over present categories(|IC_c|), -2, 2) / 2. So a filing with no ESG factors is scored using only the 13 categories that are present. There is no zero-imputation. This means filings with fewer categories have signals driven by a subset of IC weights, which could introduce noise if the subset is not representative. But we judged this preferable to imputing neutral (zero), which would bias the signal toward zero for filings with missing categories.

59. Airlines transport covers 5 tickers and 74 months. But there are 120 months in your 10-year window. That means 46 months have no airline filing data. Are those gaps because airlines did not file, or because your parser failed? If the former, how do you handle the irregular filing cadence in your backtesting — do you carry forward the last signal?

    The 74 active months reflect the actual filing cadence of the 5 airline tickers. Airlines file 10-Ks and 10-Qs on a quarterly schedule, but different airlines file at different times within a quarter, so the coverage is 74 distinct months rather than 120. We do not carry forward signals — each filing generates a one-time 21-day return observation. In months with no airline filings, there are simply no airline observations. The backtesting does not construct a continuous time-series portfolio for airlines; it computes filing-level excess returns and aggregates them across the cohort. Months with no filings contribute nothing. The 46 missing months are not parser failures; they reflect the quarterly filing schedule distributed across 5 tickers.

ROSHAN's Section (Slides 17-27)

Slide 17 — Per-Category Returns (TRAIN)

60. Your train/val/test split is 2015-2019 / 2020-2022 / 2023-2025. The validation period includes the COVID crash and recovery — arguably the most anomalous market regime in a decade. Why did you assign COVID to validation rather than keeping it in training where the model could learn from it? Did you consider an expanding-window or rolling-window approach instead of fixed splits?

    The time-based split was chosen to prevent any form of lookahead bias: training on earlier data, validating on later data, testing on the most recent data. COVID landing in the validation period is a consequence of this chronological ordering, not a deliberate choice. We considered expanding-window and rolling-window approaches but chose the fixed split for simplicity and interpretability at the midterm stage. An expanding window would be more robust — training on 2015-2017, validating on 2018, then expanding to train on 2015-2018, validate on 2019, etc. We plan to implement rolling-window backtesting for the final submission. Placing COVID in training would have let the model learn from the anomaly, but it would also risk the training signal being dominated by an extreme regime that may not recur.

61. The equal-weight average row shows VN = -1.04% and pos = +0.29%. That is a 1.33 percentage point spread. But what is the t-statistic on the difference between VN and VP mean returns in the training period? Is it significant, or could this gradient be explained by sampling noise?

    We did not compute a formal t-test on the VN vs. VP mean difference in the training period specifically. The VN cohort in training has moderate sample size (the training period contains roughly 45% of the 50 total VN filings, so about 22-23 filings), and the VP cohort in training is similarly small. With those sample sizes and the high volatility of 21-day returns (~10%), the t-statistic on the spread would likely be below 2.0. The category-level evidence is stronger: cost_margins has IS Sharpe = 1.03 with t = 2.29, computed from monthly long-short returns over the full 5-year training period. The

gradient in the equal-weight average row is directionally consistent but not conclusive on its own.

62. You report "over 27,000 factor observations" for the training period. But the unit of analysis for returns is the filing, not the factor. How many unique filings are in the training period? Each filing has one 21-day excess return, regardless of how many factors it contains. Are you double-counting returns by treating each factor-filing pair as an independent observation?

    The training period contains 1,106 unique filings. The 27,000 figure refers to factor-level observations, which we report to show the density of the data. The per-category return table uses factor-level observations appropriately: each cell shows the mean 21-day excess return for filings that had a factor in that category with that sentiment class. A filing can appear in multiple cells (e.g., in cost_margins.positive and demand_revenue.neutral) because it has multiple factors across categories. This is not double-counting within a single cell, but a filing's return does appear in multiple cells across categories. The equal-weight average row at the bottom averages across categories, not across individual factor observations, so each filing is weighted by the number of categories in which it appears. For the filing-level cohort analysis (Slide 23), each filing appears exactly once.

63. Some cells in the table show dashes, indicating zero observations. For the very_positive cohort in certain categories during training, does that mean no filing had a very positive score for that category? If so, how do you construct a long-short strategy for that category?

    Correct — dashes mean zero filings received a very_positive score in that category during that time period. This is a consequence of the base model's VP collapse: only 3.1% of all factors are VP, so in narrower category-period slices, the count can drop to zero. For the long-short strategy construction, we use positive vs. negative (not VP vs. VN), which has much denser coverage. When we report Sharpe ratios for each category, the long leg includes both positive and very_positive filings, and the short leg includes both negative and very_negative. This ensures adequate sample size for monthly return calculations. Categories with sparse extremes still contribute to the signal through the IC-weighted aggregation in Agent 4, where the category score is a continuous value, not a discrete cohort.

Slide 18 — Per-Category Returns (VAL)

64. You say the inverted gradient during COVID is "not a failure — it is a regime effect." But the entire purpose of your signal is to predict returns. If the signal inverts during market stress — precisely when risk management matters most — what is its practical value? A signal that works in calm markets but fails in crises is arguably worse than no signal at all.

    This is a fair and important criticism. A signal that inverts during crises has limited standalone value for risk management. Our defense is twofold: (1) the inversion during 2020-2022 affects the linear Spearman relationship but does not negate the tail signal — VP filings still outperform in absolute terms during this period, and (2) RL alignment in Task 3 is designed to address regime-dependence by optimizing across multiple time horizons and market conditions. Practically, a systematic strategy using this signal would need a regime detection overlay (e.g., VIX threshold, drawdown trigger) to either reduce position size or flip the signal during market stress. We have not built that overlay yet, and you are right that without it, the signal is incomplete for real portfolio

management.

65. "Companies that front-load bad news see stronger recoveries" — is this a post-hoc rationalization, or did you test it formally? Specifically, did you regress returns on sentiment interacted with a VIX or market-regime dummy to confirm the interaction effect?

    It is a post-hoc rationalization at this stage. We did not formally test the interaction between sentiment and market regime. A proper test would regress 21-day excess returns on sentiment, a regime indicator (e.g., VIX > 30 or SPY drawdown > 10%), and their interaction term. If the interaction coefficient is significant and negative, it would confirm that the sentiment-return relationship inverts during stress. This test is straightforward to implement and should be included in the final submission. Until then, the "front-loading bad news" narrative is an interpretation consistent with the data but not a validated hypothesis.

66. All cohorts show positive returns during 2020-2022. But the S&P 500 was also up dramatically. Your returns are excess versus SPY. If every cohort shows positive excess returns, does that mean the entire industrial sector outperformed SPY during the recovery? Is that a signal, or is it sector beta?

    If all cohorts show positive excess returns versus SPY during 2020-2022, it likely reflects the industrial sector's strong recovery post-COVID rather than our sentiment signal's discriminative power. The excess return is stock return minus SPY return, which removes market beta but not sector beta. If industrials as a whole outperformed SPY during this period (which they did — the re-opening trade favored cyclicals over the S&P 500's tech-heavy composition), then all cohorts would show positive excess returns regardless of sentiment quality. This is a valid criticism and supports using XLI (industrials ETF) instead of SPY as the benchmark for a more precise sector-neutral excess return. We plan to add XLI-benchmarked returns in the final submission.

Slide 19 — Per-Category Returns (TEST)

67. In the test period, most cells are negative. You say "even in a down market for the sector, the extreme positive sentiment cohorts still outperform." But "outperform" relative to what baseline? If cost_margins VP = +0.35% and the average cell is, say, -0.50%, that is outperformance of 85 bps. But is it statistically significant given the sample sizes in each cell?

    "Outperform" is relative to other cohorts within the same category — VP cells show less negative or positive returns compared to VN and negative cells in the same row. Given the small sample sizes per cell in the test period (605 filings split across 14 categories and 5 cohorts), individual cell differences are not statistically significant. The claim is directional: the VP column is consistently less negative (or positive) relative to other columns, which is consistent with the pattern from training. Statistical significance requires aggregation — either across categories (the filing-level cohort analysis) or across time (the Sharpe ratio analysis). Individual cell-level significance is not achievable with these sample sizes.

68. The test period is 2023-2025. How many filings are in the VP cohort during the test period specifically? If the overall VP count is 24 across all periods, and the split is roughly 45% train / 30% val / 25% test, you might have only 5-6 VP filings in test. Is that enough to draw any conclusion?

With 24 total VP filings across the full sample and approximately 25% in the test period, we have roughly 5-6 VP filings in the test period. That is not enough to draw any statistical conclusion about VP performance in the test period alone. Any mean return computed from 5-6 observations has an SE of approximately 10% / sqrt(6) = 4.1%, making virtually any observed mean insignificant. This is a fundamental sample-size limitation. Our test-period claims are necessarily about directional consistency with training patterns, not about statistical significance. The strongest test-period evidence comes from the category-level long-short strategies (e.g., cost_margins OOS Sharpe 0.27) which aggregate monthly returns over 3 years and have more observations.

Slide 20 — Factor Ranking IS vs OOS

69. You define the Sharpe ratio as "excess return divided by volatility, annualized." Write out the formula precisely. Is this Sharpe = (mean monthly L/S return) / (std monthly L/S return) * sqrt(12)? What is the monthly return calculation — are you equal-weighting all filings in a given month, or value-weighting?

    The formula is: Sharpe = (mean of monthly L/S returns) / (standard deviation of monthly L/S returns) * sqrt(12). Each monthly L/S return is computed as: (equal-weighted mean return of positive+VP filings that month) minus (equal-weighted mean return of negative+VN filings that month). All filings within a cohort-month are equal-weighted, not value-weighted. In months where only one side has filings (e.g., positive filings but no negative filings), we exclude that month from the calculation. This is a simple implementation — a production system would use value-weighting or risk-weighting, but equal-weighting is standard in academic factor research as a first pass.

70. Cost_margins IS Sharpe = 1.03 with t-stat = 2.29. The t-stat is Sharpe * sqrt(N_years) for an annualized Sharpe, approximately. With 5 training years, sqrt(5) * 1.03 = 2.30, which matches. But OOS Sharpe = 0.27 — that is a 74% decay. In factor research, what decay rate is considered acceptable? Is 74% decay consistent with a real signal, or with in-sample overfitting?

    In quantitative factor research, IS-to-OOS decay of 50% or more is common and does not by itself indicate overfitting. Harvey, Liu, and Zhu (2016) document that published factors typically lose 50-70% of their IS performance OOS. A 74% decay is at the high end of this range but within the bounds seen in real signals. The key question is whether the OOS Sharpe is economically meaningful and statistically significant. An OOS Sharpe of 0.27 over 6 years gives a t-statistic of approximately 0.27 * sqrt(6) = 0.66, which is not significant at conventional levels. So while the decay rate alone is not dispositive, the combination of high decay and low OOS statistical significance means we cannot confidently claim cost_margins is a robust standalone factor. It is suggestive but not conclusive.

71. Labor_workforce holds at Sharpe 0.54 OOS versus 0.58 IS — only 7% decay. Why does labor hold up so much better than cost_margins? Is there a financial explanation, or could this be luck given the small number of time-series observations (6 OOS years)?

    We do not have a definitive financial explanation. One hypothesis: labor market conditions (hiring, layoffs, wage pressure) are slow-moving fundamental signals that

persist across market regimes, whereas cost margin pressures can shift rapidly with commodity prices and supply chains. The 2020-2025 period featured persistent labor shortages and wage inflation across industrials, which may have made labor-related sentiment a consistently informative signal. However, with only 6 OOS years (72 monthly observations at most), the 7% decay could easily be luck. A Sharpe of 0.54 over 6 years gives t = 0.54 * sqrt(6) = 1.32, which is below the 1.96 threshold. So while the stability is encouraging, it is not statistically significant and could reflect a favorable regime rather than a robust signal.

72. You say "a positive out-of-sample Sharpe means the signal is real, not overfit." But with 6 OOS years of monthly data (72 observations), the t-stat for cost_margins OOS is roughly 0.27 * sqrt(6) = 0.66 — not significant. Can you really claim the signal is "real" with a t-stat below 1.0?

    No, we cannot claim statistical significance with a t-stat of 0.66. The statement in the presentation was overly strong. A positive OOS Sharpe is a necessary but not sufficient condition for a real signal. It is directionally encouraging — the signal did not reverse OOS — but it does not meet conventional significance thresholds. We should have said "the positive OOS Sharpe is directionally consistent with a real signal but not statistically significant given the limited OOS sample." This is an honest limitation of our 10-year dataset: with a 5/5 IS/OOS split (or 5/6 in our case), the OOS period is simply too short to achieve significance for factors with modest Sharpe ratios. A longer history or cross-sector validation would be needed.

73. How do you construct the monthly long-short return? If in a given month there are 3 filings with positive sentiment and 1 with negative sentiment, do you go long all three equally and short the one? What about months with no filings on either side — do you carry forward positions, sit in cash, or exclude the month?

    In a given month, we go long the equal-weighted portfolio of all filings classified as positive or very_positive in that category, and short the equal-weighted portfolio of all filings classified as negative or very_negative. If there are 3 positive and 1 negative, we equal-weight each side (each positive filing gets 1/3 weight on the long side, the 1 negative filing gets full weight on the short side). In months with no filings on either side, we exclude that month entirely from the Sharpe calculation — we do not carry forward positions or impute zero returns. In months where only one side has filings (e.g., 3 positive, 0 negative), we also exclude the month since a long-short return requires both legs. This means the effective number of monthly observations may be less than the full 60 (training) or 72 (OOS) months.

Slide 21 — Cumulative Returns

74. The cumulative return chart for cost_margins shows steady upward drift. But is this a cumulative *sum* of monthly returns or a cumulative *product* (compounded)? The distinction matters for interpreting the economic magnitude. Also, is this gross or net of transaction costs?

    The cumulative return chart shows a cumulative sum of monthly long-short returns, not a compounded product. This is the standard presentation in factor research because it provides a clearer visual of the monthly return stream. A compounded product would

show slightly higher terminal values due to compounding effects. The returns are gross of transaction costs — we have not modeled trading costs, slippage, or market impact. Given the quarterly filing cadence and 21-day holding period, turnover is moderate (roughly 4 turns per year per leg), so transaction costs would be material but likely not prohibitive. At 10 bps round-trip, annual trading costs would be approximately 80 bps, which would reduce the cumulative return but not eliminate it.

75. You say cost_margins "keeps climbing through the validation boundary and through the test boundary." But I notice the equal-weight average peaks around 2020 and declines. That means most of your 14 factors are failing OOS. Aren't you cherry-picking cost_margins because it is the one survivor? How many of the 14 categories would you expect to show positive OOS drift by random chance alone?

    Under the null hypothesis of no signal, with 14 independent tests, you would expect approximately 7 to show positive OOS drift and 7 negative, purely by chance. So observing 1 strong OOS survivor out of 14 is not statistically impressive — it is well within the range of random chance. You are right to flag this as potential selection bias. The fact that cost_margins was the top IS factor and also survives OOS is more encouraging than picking a random OOS survivor, because it was pre-specified (ranked #1 IS). But formally, the probability that the best IS factor also shows positive OOS drift by chance is roughly 50%, not 5%. We should apply a multiple-testing correction (e.g., Bonferroni or FDR) to properly assess whether any factor survives at conventional significance levels after accounting for the 14 hypotheses tested.

76. Is there a look-ahead bias in selecting cost_margins as the "top" factor? You identified it as the best in-sample factor and then checked it OOS. But if you had 14 factors and tested all of them, the probability that at least one shows a positive OOS Sharpe by chance is much higher than the probability for any single pre-specified factor. Did you apply a Bonferroni or false-discovery-rate correction?

    We did not apply a Bonferroni or FDR correction, which is a valid criticism. With 14 factors, the Bonferroni-adjusted significance threshold at alpha = 0.05 would be 0.05/14 = 0.0036, requiring a t-stat of approximately 2.9. None of our OOS Sharpe ratios meet this bar. The probability that at least one of 14 factors shows a positive OOS Sharpe by chance (assuming independence and null hypothesis) is 1 - 0.5ˆ14 = 99.99%, so finding positive OOS Sharpes is essentially guaranteed by chance alone. The more relevant test is whether the OOS Sharpes are significantly positive after adjustment. They are not. We should present the multiple-testing-corrected results in the final submission for intellectual honesty.

Slide 22 — Rolling t-Statistics

77. You show 12-month and 24-month rolling t-statistics. The 12-month window is very short — with monthly data, you have only 12 observations per window, and a t-test with 11 degrees of freedom requires t > 2.20 for 95% significance (not 1.96). Are your significance lines correctly drawn?

    You are correct — for a 12-month rolling window with 11 degrees of freedom, the 95% significance threshold is approximately 2.20 (two-tailed), not 1.96. We drew the significance lines at 1.96, which is the asymptotic (large-sample) critical value. This is a

minor but real error that makes the chart appear to show significance more frequently than it actually does. For the 24-month window with 23 degrees of freedom, the correct threshold is approximately 2.07, which is closer to 1.96 but still higher. We should correct the significance lines in the final version.

78. Cost_margins "crosses above 1.96 multiple times." How many months total does it spend above 1.96 versus below? If it crosses above briefly and spends most of the time below, the visual impression of persistent significance is misleading.

    We did not compute the exact fraction of months above vs. below the significance threshold. The visual impression from the chart is that cost_margins spends the majority of time below the significance line and crosses above during specific periods (mid-training, early validation, and briefly in test). The periods above are episodes, not persistent states. This means the signal's significance is time-varying and regime-dependent, not consistently strong. The visual impression of "persistent significance" may indeed be misleading if most of the 120-month window is spent below the threshold. For the final submission, we should report the exact fraction of months above significance for each window length.

79. Rolling t-statistics are serially correlated by construction — each window overlaps the previous by 11 (or 23) months. Does this affect your interpretation? A factor could appear "stable" simply because the rolling window carries the same few strong observations forward.

    Correct — rolling windows with overlapping observations produce serially correlated t-statistics by construction. A single strong month can inflate the t-statistic for 12 (or 24) consecutive rolling windows, creating an illusion of persistence. This does affect interpretation: we cannot treat each rolling window as an independent test. The visual continuity of the t-statistic line overstates the true stability of the signal. A more rigorous approach would use non-overlapping windows (e.g., 6-month non-overlapping blocks) or compute block bootstrap confidence bands. We should add non-overlapping window analysis in the final submission to provide a less biased view of signal stability.

Slide 23 — Cohort vs. 21-Day Excess Return (THE KEY CHART)

80. This is the chart you call "the most important." Let me stress-test it. VN mean = +0.91% (n=50). The standard error on that mean — assuming, say, 10% return volatility over 21 days — is approximately 10% / sqrt(50) = 1.41%. So the 95% confidence interval for VN is roughly [-1.87%, +3.69%]. The VP mean of +3.69% (n=24) has SE = 10% / sqrt(24) = 2.04%, giving a 95% CI of [-0.31%, +7.69%]. These confidence intervals overlap massively. Can you really distinguish VP from VN statistically?

    No, we cannot statistically distinguish VP from VN at the 95% level. The confidence intervals overlap significantly, as you computed. A two-sample t-test on VP (mean 3.69%, n=24) vs. VN (mean 0.91%, n=50) gives a mean difference of 2.78% with a pooled SE of approximately sqrt(2.04%^2 + 1.41%^2) = 2.48%. The t-statistic is 2.78% / 2.48% = 1.12, well below 1.96. So the difference is not significant at the 95% level. We are transparent about this: the VP tail is directionally strong but statistically inconclusive due to small sample size. The converging evidence from category-level analysis (cost_margins t = 2.29) provides complementary support, but the headline cohort chart alone does not pass a formal significance test.

81. The Spearman correlation is 0.009 with p = 0.655. That is indistinguishable from zero. You acknowledge this. But then you pivot to "the +3.69% tail signal tells us the pipeline is extracting real information." How do you reconcile a null Spearman with claiming the signal works? Is it not possible that the VP result is simply a small-sample anomaly?

    It is possible. The null Spearman means there is no monotonic rank-order relationship across all five cohorts. The +3.69% VP tail could be a small-sample anomaly — we cannot rule that out with n=24. Our reconciliation is that Spearman measures the full-distribution rank correlation, which is dominated by the 1,314 neutral filings (54%) that carry no signal. The tails (74 filings, 3%) are overwhelmed statistically. A U-shaped or threshold-based relationship — where only the extremes are informative — would show a near-zero Spearman even if the tails contain real signal. But we acknowledge this is a weaker claim than "the signal works across the distribution." The honest statement is: we see suggestive tail performance that is consistent with category-level evidence, but the overall signal is not statistically significant. RL alignment is designed to improve this.

82. The long-short spread is VP minus VN = +2.78%. But the relationship is not monotonic — VN actually outperforms neutral by 58 basis points. If you built a portfolio based on this signal and went short VN, you would *lose* money on the short leg. Is this really an investable signal?

    As a simple long VP / short VN strategy, no — you would not lose money on the short leg (VN returns are positive at +0.91%, so shorting VN would cost -0.91%), and the long leg earns +3.69%, netting +2.78%. But the non-monotonicity is a problem. A well-behaved signal should show VN < negative < neutral < positive < VP. The fact that VN (+0.91%) outperforms negative (+0.12%) and is close to neutral (+0.33%) means the short leg does not add value — it actually hurts you. A more practical strategy would be long-only VP rather than long-short VP/VN. The non-monotonicity is precisely what RL alignment (Task 3) is designed to fix: GRPO will optimize the model's output distribution to enforce a monotonic relationship between sentiment cohorts and subsequent returns.

83. You report IR = 0.067. Information ratio is typically defined as the mean active return divided by tracking error. Over what period — the full sample, or a specific sub-period? With 2,441 filings, what is the annualized long-short return and its volatility?

    The IR of 0.067 is computed over the full sample (2015-2025). It is defined as the annualized mean L/S return divided by the annualized standard deviation of L/S returns (tracking error). With 2,441 filings across ~120 months, the monthly L/S return has a very small mean (the VP-VN spread of 2.78% is on a per-filing basis, not per-month) and moderate volatility. An IR of 0.067 is low — institutional standards typically require IR > 0.5 for deployment. This reflects the fact that the continuous signal has low discriminative power across the full distribution. The IR would be higher if computed only on the tail portfolios (top and bottom deciles) rather than the full 5-cohort L/S.

84. The 21-day excess return is computed as stock return minus SPY return. But SPY is a market-cap-weighted index dominated by mega-cap tech. For an industrials-focused universe, would subtracting XLI (industrials ETF) or using a Fama-French industry factor adjustment be more appropriate? Your "excess return" includes sector beta relative to SPY.

You are correct. SPY is dominated by mega-cap tech (Apple, Microsoft, Nvidia comprise ~20% of SPY), making it an imperfect benchmark for an industrials-only universe. Our "excess return" conflates two sources: (1) alpha from the sentiment signal and (2) time-varying sector beta (industrials vs. tech-heavy SPY). Subtracting XLI would produce a cleaner sector-neutral excess return. A Fama-French industry factor adjustment or a CAPM regression with an industrials benchmark would be even more rigorous. We used SPY because the provided returns data (filing_returns_1m.csv) was pre-computed against SPY. For the final submission, we plan to compute XLI-adjusted returns to isolate whether our signal provides alpha above and beyond sector membership.

85. n=24 for VP. How many of those 24 are from the same ticker? If 10 of the 24 VP filings are from, say, Caterpillar during a strong earnings run, you are not measuring signal quality — you are measuring Caterpillar's stock performance. What is the ticker concentration in the VP cohort?

    We did not report the ticker concentration within the VP cohort, which is an important diagnostic we should have included. If the 24 VP filings are concentrated in a few tickers, the excess return reflects company-specific performance rather than signal quality. A proper test would compute the effective number of independent observations (adjusting for within-ticker correlation) and verify that no single ticker dominates the cohort. For the final submission, we will report the ticker distribution within each cohort and compute cluster-robust standard errors that account for within-ticker dependence. If ticker concentration is high, we should also report the median VP return (which is more robust to individual ticker dominance) alongside the mean.

86. You present this chart as the output of the full pipeline. But the SFT model was not used to re-score the 67K factors — Agent 1 uses base model scores. So this chart actually represents the base model pipeline, not the SFT-enhanced pipeline. Is that correct? If so, why does the title card say "SFT-only pipeline"?

    That is correct — the chart uses base model scores, not SFT-rescored factors. The "SFT-only pipeline" label in the presentation refers to the pipeline stage (post-SFT, pre-RL) rather than the scores used. This labeling is misleading and we should have said "base model pipeline" or "pre-RL pipeline." The SFT model was trained and validated but not applied to re-score the full corpus. The chart represents what the base model + multi-agent consolidation produces. The SFT-rescored version is planned but not yet executed.

Slide 24 — Signal vs. 21-Day Excess Return

87. The OLS regression line is flat. What is the slope coefficient, its standard error, and its p-value? If the slope is not significantly different from zero, then the continuous signal has no linear predictive power. How does that square with your claim that "the pipeline's value is in the tails"?

    The OLS slope is not significantly different from zero — the overall Spearman of 0.009 ($p = 0.655$) confirms the absence of a monotonic relationship, and OLS would show a similarly null result. The slope would be approximately 0.1-0.3% per unit of signal, with a standard error large enough to include zero. This is fully consistent with our claim that "the pipeline's value is in the tails": a linear model fits the conditional mean across

the full distribution, which is dominated by the 54% of filings in the neutral cohort where the signal carries no information. A linear model cannot capture threshold or step-function relationships where only the extremes matter. A more appropriate model would be a piecewise regression or a quantile regression focused on the tails.

88. The return cloud spans +/- 20%. Individual stock returns over 21 days have enormous idiosyncratic volatility. Given this noise level, what signal-to-noise ratio would you need for the continuous signal to achieve statistical significance? How many filings would be required?

    With 21-day return volatility of approximately 10% (standard deviation), and a signal effect of approximately 1-2% per unit of signal, the signal-to-noise ratio is roughly 0.1-0.2. For a regression slope to achieve t > 1.96, we need n > (1.96 * sigma / effect)ˆ2. With sigma = 10% and effect = 1%, that is n > (1.96 * 10)ˆ2 = 384 filings at a minimum, assuming the signal is truly linear and uniformly distributed across signal values. We have 2,441 filings, which should be sufficient if the linear relationship existed. The fact that it does not appear with 2,441 observations strongly suggests the relationship is not linear — it is concentrated in the tails, as we claim. A larger universe (more sectors, more tickers) would provide more VP and VN observations, which is where the real test lies.

89. You suggest Task 3 GRPO will "push the model to produce sharper, more extreme predictions." But if the continuous signal is not predictive, training RL to maximize return prediction on a noisy target is prone to overfitting. What regularization or reward-shaping mechanisms do you plan to prevent the RL policy from fitting noise?

    This is a genuine risk. Our planned mitigations include: (1) KL divergence penalty against the SFT model to prevent the RL policy from drifting too far from the pretrained distribution, (2) group relative policy optimization (GRPO) rather than PPO, which compares outputs within a group rather than against an absolute reward, reducing variance, (3) reward shaping that incorporates monotonicity as a constraint — not just maximizing return prediction, but penalizing non-monotonic cohort-return relationships, and (4) training on the training period (2015-2019) and validating on 2020-2022 to detect overfitting. We also plan to use the actual stock return as a noisy reward signal rather than trying to predict returns directly, which means the RL objective is to improve the rank ordering of sentiment labels relative to returns, not to predict return magnitudes.

Slide 25 — Per-Sector Cohort vs. Return

90. Airlines VP outlier is ~13% return with n=1. You "set it aside," but it is included in the aggregate VP mean of +3.69%. If you remove that single airline VP filing, what happens to the overall VP mean? Could one outlier be driving your headline result?

    With n=24 and a mean of +3.69%, the total sum of VP returns is 24 * 3.69% = 88.56%. If the airline outlier contributes 13%, removing it gives a sum of 75.56% across 23 filings, or a mean of 75.56% / 23 = 3.29%. So the VP mean drops from +3.69% to +3.29% — a reduction of 40 bps. The headline result is not solely driven by this one outlier; the remaining 23 filings still average +3.29%. However, this illustrates the fragility of small-sample means: any single observation in a n=24 cohort can move the mean by ~40-50 bps. Reporting the median VP return alongside the mean would provide a more robust measure that is less sensitive to outliers.

91. Defense Spearman = -0.148 (p=0.007). You interpret this as a contrarian signal. But in a long-short context, you would need to go *long* negative-sentiment defense filings and *short* positive-sentiment ones. That is the opposite direction from the rest of your portfolio. How do you handle this in portfolio construction — do you flip the signal for defense, or exclude the sector?

    In the current pipeline, we do not flip the signal for defense or exclude the sector. The IC-weighted aggregation in Agent 4 uses the absolute value of IC (|IC_c|), which means defense categories contribute to the signal magnitude but not in a sign-flipped manner. This is a design gap. In proper portfolio construction, there are two approaches: (1) flip the sign of defense category scores before aggregation, effectively treating negative defense sentiment as positive, or (2) exclude defense from the IC-weighted signal and trade defense as a separate contrarian strategy. Approach (1) is simpler but assumes the contrarian relationship is stable, which we have not verified OOS. For the final submission, we should evaluate both approaches and compare their impact on portfolio-level Sharpe.

92. General and Industrial Equipment together show Spearman of 0.039 and 0.018 — both insignificant. These two sectors comprise 1,918 of your 2,441 filings (78%). So for the overwhelming majority of your universe, there is no statistically significant relationship between your signal and returns. How do you reconcile this with claiming the overall pipeline works?

    We reconcile it by acknowledging that the filing-level Spearman across the full distribution is not our primary evidence. The raw filing-level signal is noisy — 54% of filings fall in the neutral cohort with no signal content. The evidence for the pipeline comes from: (1) category-level factor analysis, where cost_margins achieves IS Sharpe 1.03 (t = 2.29, significant) and positive OOS Sharpe, (2) the VP tail showing +3.69% excess return that is directionally strong though not individually significant, and (3) the IC-weighted aggregation that concentrates signal from the predictive categories. We are honest that the overall linear relationship is null. The pipeline's value proposition is extracting signal from specific categories and extreme tails, not from the average filing. Whether this is sufficient for a real investment strategy is an open question that the final submission should address with proper multiple-testing corrections and net-of-cost analysis.

93. The per-sector scatter plots show considerable overlap between cohorts. Within the General sector (n=837), what is the VP cohort size? If it is single digits, the per-sector VP mean is meaningless.

    The VP cohort within the General sector is likely in single digits. With 24 total VP filings across all sectors and General comprising ~34% of the universe, we would expect approximately 8 VP filings in General. With n=8, any mean return is statistically meaningless — the SE would be approximately 10% / sqrt(8) = 3.5%, producing a confidence interval that spans from deeply negative to strongly positive. We should not draw per-sector VP conclusions with these sample sizes. The per-sector scatter plots are useful for visualizing the overall distribution and the Spearman correlation but are not informative about cohort-level performance when the extreme cohorts contain single-digit observations.

Slide 26 — Next Steps

94. You plan GRPO across 1-month, 3-month, and 6-month horizons. GRPO requires a reward signal. What exactly will the reward function be — the raw excess return at each horizon, a discretized quintile rank, or something else? How will you handle the fact that 3-month and 6-month returns overlap temporally, creating correlated rewards across training samples?

    The planned reward function will use the raw excess return at each horizon, with monotonicity penalty: reward = alpha * excess_return * sign(sentiment_score) + beta * monotonicity_bonus, where the monotonicity bonus rewards the model when cohort-level mean returns are monotonically ordered. For the temporal overlap issue with 3-month and 6-month returns, we plan to either (1) train separate models for each horizon and ensemble, or (2) use multi-objective GRPO with weighted rewards across horizons, down-weighting correlated samples. The specific implementation is still in design phase. Overlapping returns create correlated training samples, which reduces the effective sample size and can lead to overfitting. We will need to account for this in the reward variance estimation within GRPO's relative policy gradient.

95. You mention ORM and PRM as a "bonus track." Given that your primary signal is not yet significant, would the team's time be better spent on improving the base signal (e.g., re-scoring with SFT, trying alternative return horizons, sector-specific models) rather than adding architectural complexity with ORM/PRM?

    You are right — our priority should be strengthening the base signal before adding complexity. The most impactful next steps are: (1) re-scoring the full 67K corpus with the SFT model and re-running the backtest, (2) computing XLI-adjusted returns to isolate sector-neutral alpha, (3) testing alternative return horizons (5-day, 10-day, 63-day), and (4) applying multiple-testing corrections to our factor analysis. ORM/PRM is listed as a bonus because it is genuinely a stretch goal, contingent on the core pipeline demonstrating significant signal first. If GRPO alone does not produce a monotonic cohort-return curve, adding ORM/PRM reasoning supervision is unlikely to solve the underlying data problem.

96. For portfolio construction, do you plan to account for transaction costs? With 21-day holding periods and quarterly filing cadence, turnover could be substantial. At, say, 10 bps round-trip, what fraction of the +2.78% long-short spread would be consumed by trading costs?

    With quarterly filing cadence (roughly 4 filings per ticker per year), each filing generates one trade entry and one exit over 21 days. For a portfolio holding ~20-30 positions at a time across both legs, annual turnover is roughly 4x per position. At 10 bps round-trip per trade, the annual trading cost would be approximately 4 * 10 bps = 40 bps per leg, or 80 bps total for the long-short portfolio. The +2.78% L/S spread is a per-filing figure, not an annualized return. To properly assess the cost impact, we need the annualized L/S return, which depends on how many VP and VN filings occur per year. With ~24 VP filings over 10 years (2.4/year), the trading frequency is very low, and transaction costs would consume a small fraction of the per-trade alpha. We will include a formal transaction cost model in the final submission.

97. You say "RL enforces monotonic cohort-return relationship." But monotonicity is a property of the empirical return distribution conditional on the signal — it is not something you can directly optimize. GRPO optimizes the model's output policy. How specifically does the

reward function encode monotonicity? Is it reward = excess_return * sign(sentiment), or something more nuanced?

The reward function encodes monotonicity indirectly. GRPO optimizes the model's sentiment output policy by rewarding outputs that, when aggregated across a batch of filings, produce a monotonic mapping from sentiment to returns. The planned approach: for each batch, compute the mean return per cohort, then reward the model proportionally to how well the cohort means are ordered (e.g., using Spearman rank correlation between cohort order and mean return as a batch-level reward signal). The per-sample reward would be: reward_i = excess_return_i * numeric_sentiment_i (rewarding agreement between sentiment direction and return direction) plus a batch-level monotonicity bonus distributed across samples. This is more nuanced than simple reward = return * sign(sentiment), but the exact formulation is still being finalized.

Slide 27 — Thank You / Summary

98. You list "VP F1 0.206 to 0.793" as a key achievement. But this improved F1 is measured on the validation set where the labels come from the base model itself. The improvement means the SFT model better reproduces the base model's high-confidence very_positive labels. It does not mean the very_positive labels are *correct* in any absolute financial sense. How would you validate that VP labels correspond to genuinely positive financial outcomes?

    The only way to validate VP labels against financial reality is to test whether VP-labeled factors predict positive excess returns — which requires the SFT-rescored backtest we have not yet run. The VP F1 improvement (0.206 to 0.793) measures consistency with the base model's high-confidence labels, not financial accuracy. The base model's VP labels could be systematically wrong — e.g., labeling management optimism as "very positive" when the underlying fundamentals are deteriorating. The circumstantial evidence that VP labels have financial content comes from the base model pipeline's +3.69% VP cohort return, but this uses base model scores, not SFT scores. The definitive test is: after SFT re-scoring, does the VP cohort still outperform, and does it outperform more than the base model VP cohort? If not, the F1 improvement was cosmetic.

99. "Cost_margins IS Sharpe 1.03 OOS 0.27" — you present this as evidence of robustness. But in institutional quant research, a factor with OOS Sharpe of 0.27 after only 6 years would not meet the bar for deployment. What Sharpe would AllianceBernstein need to see to consider this tradeable?

    Institutional quant firms typically require an annualized Sharpe of 0.5 or higher for a standalone factor to be considered for deployment, with a t-statistic of at least 2.0 over a meaningful OOS period (typically 5+ years). Our OOS Sharpe of 0.27 with t = 0.66 does not meet this bar. In practice, most deployed factors have ISharpe > 1.5 and OOS Sharpe > 0.7, with evidence across multiple asset classes or geographies. Our cost_margins factor would need to approximately double its OOS Sharpe and be validated across a longer time period or broader universe. That said, factors are rarely deployed in isolation — a multi-factor portfolio combining cost_margins with labor_workforce and other signals could achieve a higher portfolio-level Sharpe through diversification. Our result is a research-stage finding, not a deployment-ready signal.

100. Looking at the overall project: you have 2,441 signals, of which only 24 are VP. Your headline

claim rests on those 24 observations. Your Spearman across the full sample is 0.009 (null). Your continuous signal shows a flat OLS line. The SFT model was not actually used in the backtested pipeline. And the "multi-agent system" is a rule-based weighted average. Given all of this, what is your single strongest piece of evidence that this pipeline produces a genuine investment signal, as opposed to noise decorated with an impressive infrastructure?

Our single strongest piece of evidence is the cost_margins category achieving an in-sample Sharpe of 1.03 with a t-statistic of 2.29 (significant at 95%) on a pre-specified long-short strategy over 2015-2019, followed by a positive (though not individually significant) OOS Sharpe of 0.27 over 2020-2025. This is stronger evidence than the VP cohort because: (1) it is computed from monthly returns over 60+ months, not 24 point observations, (2) it was the top-ranked IS factor and retained its sign OOS, and (3) it uses a well-defined financial category (cost and margin sentiment) with a clear economic rationale — companies flagged for margin pressure underperform. We acknowledge all the limitations you list. The honest characterization of our midterm result is: we have built infrastructure that extracts structured financial features from raw filings, and we see suggestive but not conclusive evidence of predictive power concentrated in specific categories and extreme tails. The remaining work — SFT re-scoring, RL alignment, proper multiple-testing correction, and sector-neutral benchmarking — will determine whether this is real signal or noise.

Cross-Cutting / General Questions

101. The project spec's "Expected Performance" chart shows SFT+RL achieving a monotonic cohort-return curve with Long-Short return of roughly 2.0-2.5%. Your pre-RL Long-Short is already +2.78%. Are you already ahead of the spec's expected SFT+RL performance? If so, does that suggest the expected performance chart was conservative, or that your result is overstated due to small-sample effects?

    Most likely our result is overstated due to small-sample effects. The +2.78% L/S spread comes from VP (n=24) minus VN (n=50), and the confidence intervals on both means are wide enough to include zero. The spec's expected 2.0-2.5% with monotonic ordering across all five cohorts is arguably a harder target than our non-monotonic +2.78% spread between just the two extremes. Our spread looks numerically large because the VP mean is inflated by small-sample variance. With 24 observations, a few strong returns can easily push the mean above 3%. As n increases (through SFT re-scoring expanding the VP cohort or through RL sharpening predictions), the mean will likely regress toward a more modest value. Being "ahead" of the spec on raw spread while being behind on monotonicity and statistical significance is not a genuine lead.

102. Have you considered the filing-date timing issue? Filings appear on EDGAR after market close, and you measure returns "starting the day after the filing." But institutional investors and NLP-based trading systems parse filings within minutes of release. By the next day's open, much of the information may be priced in. Are your excess returns measured from close-to-close, open-to-close, or something else?

    Our excess returns use the filing_returns_1m.csv data, which computes 21-day returns starting from the trading day after the filing date. The exact timing convention (close-to-close vs. open-to-close) is inherited from the provided data and we believe it is close-to-close. You raise a valid point: fast NLP systems and institutional investors can

parse filings within minutes of EDGAR publication, so the Day+1 open may already reflect much of the information content. Any alpha we observe is net of this first-mover disadvantage. The fact that we still see some tail signal suggests either (1) the market does not fully price the specific factor-level information our pipeline extracts, (2) the signal operates over a longer horizon than intraday, or (3) it is noise. Testing Day+0 (same-day) returns versus Day+1 returns would help quantify how much alpha is consumed by the overnight gap.

103. You are all in the industrials sector. The period 2015-2019 includes the Trump tariff escalation, 2020-2022 includes COVID and supply chain disruptions, 2023-2025 includes the AI capex cycle. These are massive macro regimes. Is your signal genuinely extracted from filing text, or is it a proxy for macro sentiment that could be captured more simply by, say, PMI or ISM manufacturing data?

    This is a critical question we have not formally tested. Our factor categories include macro_external (with questions about economic conditions, interest rates, and geopolitical risk) which could directly proxy for PMI/ISM. If our signal is primarily capturing macro sentiment, then a simple ISM-based trading rule would dominate our expensive LLM pipeline. The proper test is to regress our filing-level signal on contemporaneous macro indicators (PMI, ISM, VIX, yield curve slope) and check whether the residual still predicts returns. If the macro-adjusted signal loses all predictive power, our pipeline is an expensive macro proxy. If it retains predictive power, the pipeline extracts firm-specific information beyond macro conditions. We should include this macro-orthogonalization test in the final submission. Notably, cost_margins and labor_workforce — our best-performing categories — are firm-specific and less likely to be pure macro proxies than macro_external or regulatory_legal.

104. What is the latency of your pipeline? From the moment a filing appears on EDGAR, how long until you produce a filing-level signal? If it takes hours to run the LLM pipeline, the signal may be stale before you can trade on it. What is the practical implementation path for a real trading desk?

    End-to-end latency for a single filing is approximately 15-30 minutes: HTML parsing (~1 minute), subsection splitting (~1 minute), factor extraction with chain-of-thought (~10-20 minutes for 20 subsections at 60 questions), sentiment scoring (~5 minutes for 25-30 factors), and multi-agent consolidation (~seconds). This is too slow for intraday event-driven trading, where signals are stale within minutes. For a real trading desk, the practical path would be: (1) reduce inference latency by using a faster model (e.g., 7B parameter) or optimized serving (TensorRT, speculative decoding), targeting <5 minutes total, (2) accept that the signal is not intraday-competitive and instead use it for next-day or weekly portfolio rebalancing, (3) pre-compute question embeddings and subsection routing to eliminate redundant processing. Our 21-day holding period means a 30-minute delay is economically irrelevant — the signal value unfolds over weeks, not minutes.

105. You have not discussed risk at all — drawdowns, maximum loss, worst-case scenarios. What is the worst single-filing outcome in the VP cohort? If a VP-labeled filing subsequently drops 20%, that wipes out multiple winning trades. What is the skewness and kurtosis of returns within each cohort?

    We did not report risk metrics, which is a significant gap. The worst single-filing

outcome in the VP cohort could be as bad as -15% to -20% over 21 days — individual stock returns have heavy left tails from earnings misses, guidance cuts, or sector-wide sell-offs. With n=24 VP filings, one -20% observation would reduce the VP mean by approximately 100 bps (from +3.69% to ~+2.7%). We did not compute skewness or kurtosis within each cohort. For a complete risk analysis, the final submission should include: (1) max drawdown of the cumulative L/S strategy, (2) worst single-month return, (3) return distribution statistics (skewness, kurtosis) by cohort, (4) Value-at-Risk (VaR) estimates, and (5) conditional tail expectations. Without these risk metrics, presenting the return results gives an incomplete and potentially misleading picture of the strategy's attractiveness.